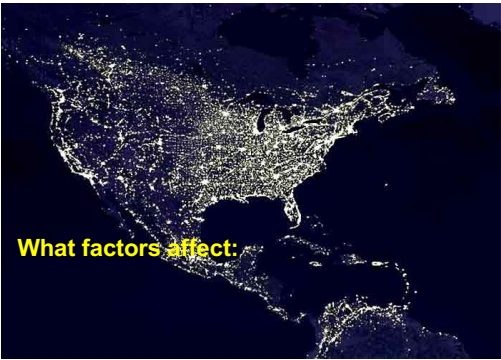


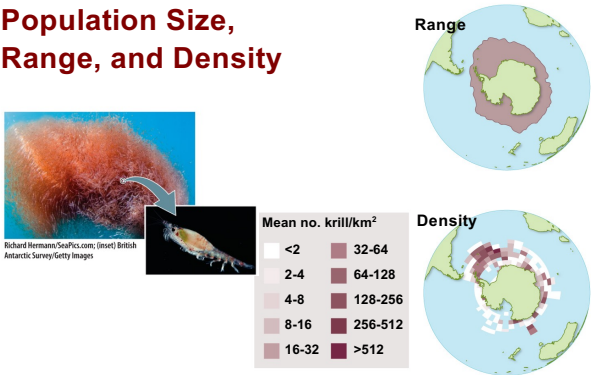
Chapter 51 Objectives: Be able to...

- Describe several factors that influence the **size, range, and density** of a population.
- Read & interpret a **life table**, and explain how they can be used as a tool to help **conserve species**.
- Describe **fitness trade-offs** and different **life history strategies**.
- Use the **logistic growth equation** to calculate changes in a population.
- Provide examples of **density-dependent** and – **independent factors** that limit populations.
- Describe **metapopulation dynamics** and how they relate to conservation.

Population:

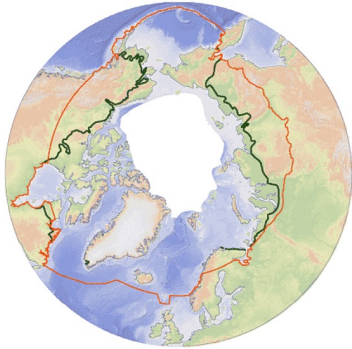


Population Size, Range, and Density

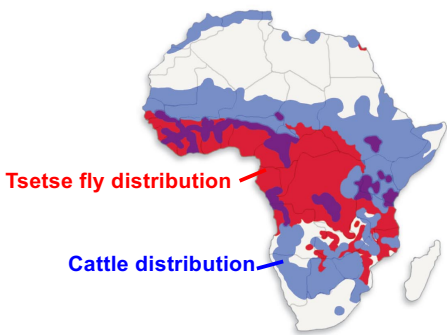


Biogeography: Geographic distribution of organisms

Species **range** limited by:



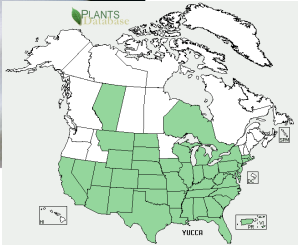
Species range limited by:



Species range limited by:



Yucca moths require yucca flowers

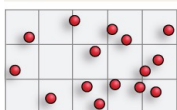


Species range limited by:

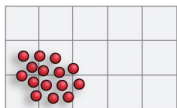




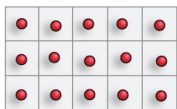
Distribution of individuals within a population



Random



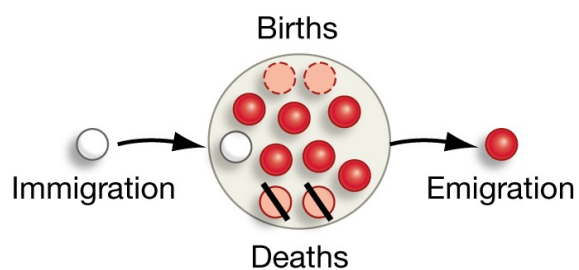
Clustered



Uniform

Table 51.1

Factors that change population size:



Per Capita Growth Rate
(Ignoring Immigration and Emigration)

$$r = \frac{\Delta N / \Delta t}{N_1}$$

Predicting Population Size
(Ignoring Immigration and Emigration)

$$N_t = N_1 + r N_1$$

Exponential Growth Occurs at Constant r

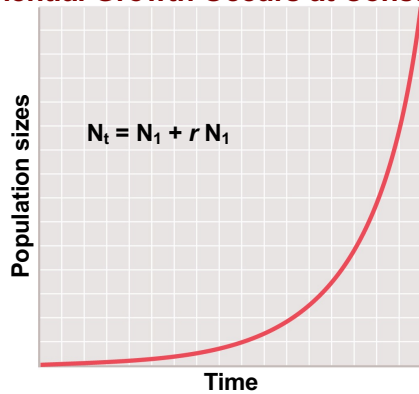


Fig 44.5

Density-dependent growth: In real life, there are limits...

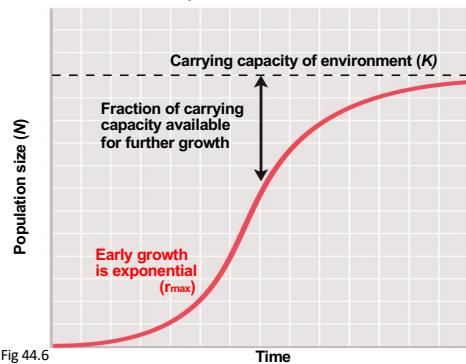


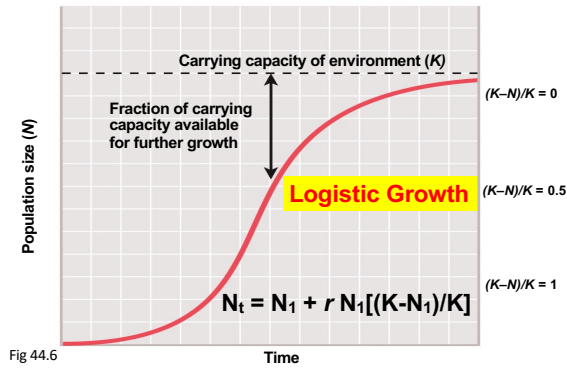
Fig 44.6

Exponential growth equation: $N_t = N_1 + r N_1$

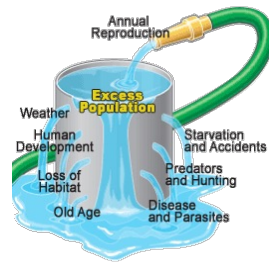
Logistic growth equation:

$$N_t = N_1 + r N_1 \left(\frac{K - N_1}{K} \right)$$

Density-dependent growth:
In real life, there are limits...



Carrying Capacity:



Life Table:



Age Class (x)	Number of Survivors (N_x)	Survivorship (l_x)	x
0	1000	1.000	
1	424	0.424	
2	308	0.308	
3	158	0.158	
4	57	0.057	
5	10	0.010	
6	7	0.007	
7	2	0.002	

Table 51.2

Age Class (x)	Number of Survivors (l_x)	Survivorship (l_x)	x	Age-Specific Fecundity (m_x)	=	Average Births/Year/Original Female ($l_x m_x$)
0	1000	1.000		0.00		0.00
1	424	0.424		0.04		0.02
2	308	0.308		1.47		0.45
3	158	0.158		2.07		0.33
4	57	0.057		2.44		0.14
5	10	0.010		3.25		0.03
6	7	0.007		3.25		0.02
7	2	0.002		3.25		+ 0.01

$R_0 = 1.00$ = Net reproductive rate

Things you should get from a life table:

Table 51.2

Survivorship Curves: Type I

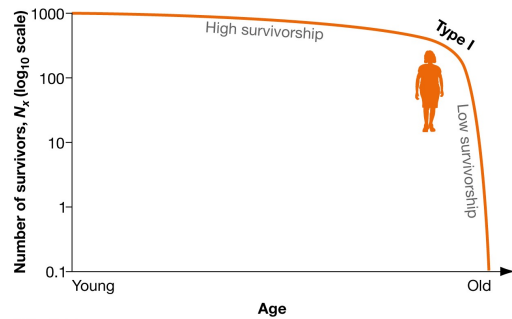


Fig. 51.3

Survivorship Curves: Type II

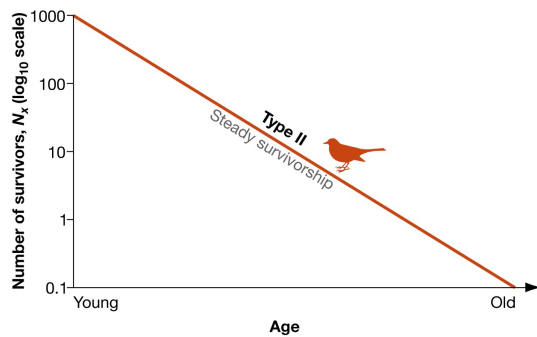


Fig. 51.3

Survivorship Curves: Type III

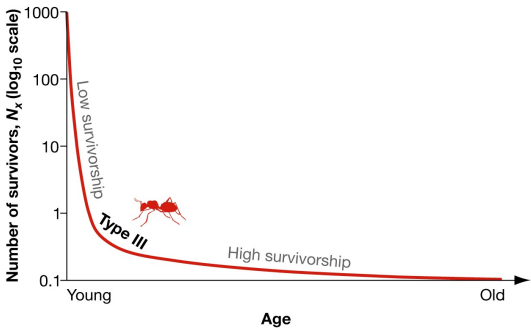
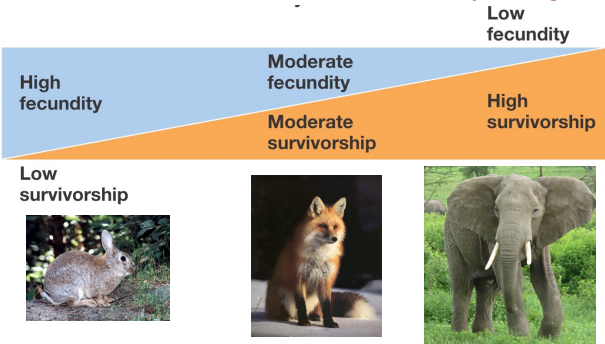
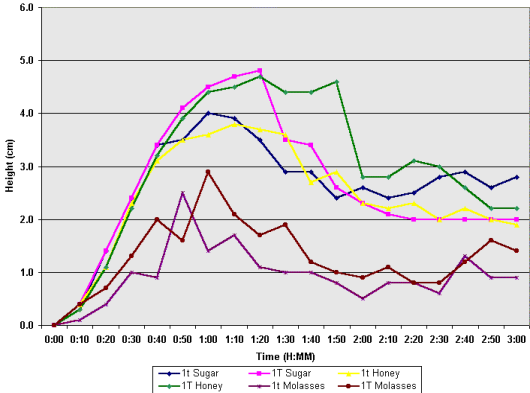


Fig. 51.3

Fitness trade-offs lead to different life history strategies



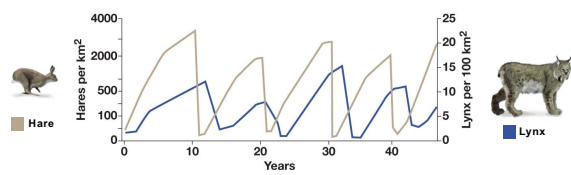
Yeast growth in different sugars



Density-dependent
factors

Density-independent
factors

Population cycles



Why do hare populations crash?

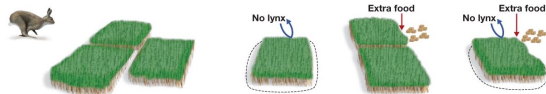
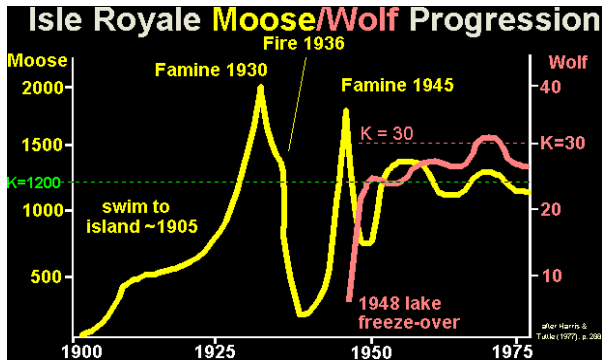
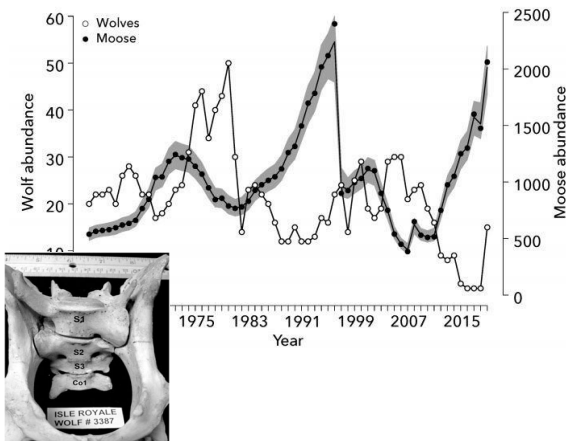


Fig. 51.10

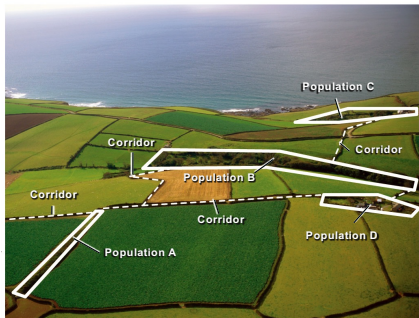




Predators often stabilize prey populations



Metapopulations:



Metapopulations: Isolated vs. Connected Patches

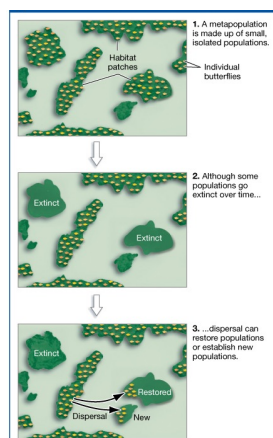
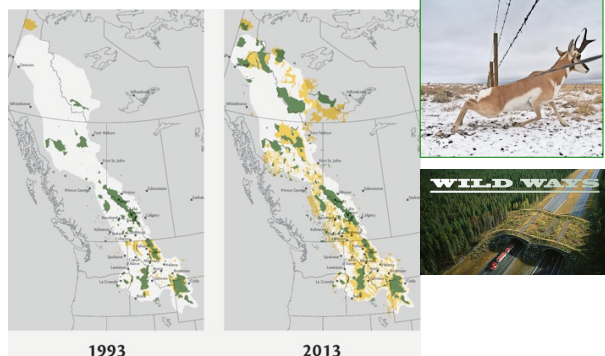


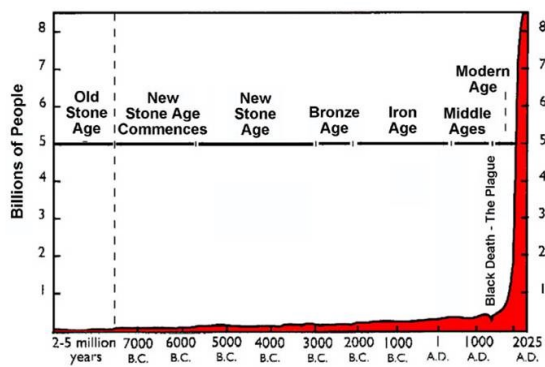
Fig 51.12

Metapopulation Dynamics are Important in Conservation



Yellowstone To Yukon Corridor





Current World Population

Human population with various fertility rates

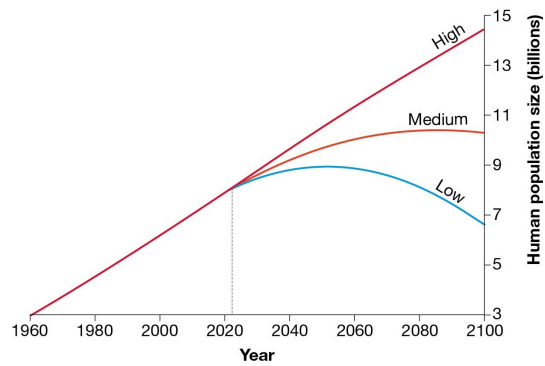


Fig 51.16

TABLE 52.2 Milestones in Human Population Growth

Date	Human Population (in hundreds of millions)
1420	375
1720	750
1800	1000
1875	1500
1927	2000
1961	3000
1974	4000
1987	5000
1999	6000
2011	7000
2022	8000